

Nemanja Bartolovic

COMPUTER GRAPHICS SOFTWARE ENGINEER

Zurich, Switzerland

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Work Experience

Senior Staff Software Engineer

Zurich, Switzerland

META PLATFORMS, REALITY LABS

August 2021 - Present

- Tech lead for real-time neural graphics and novel representations, coordinating work across multiple teams and organizations to ship research ideas into production.
- Spearheaded a company-wide, unified photorealistic real-time renderer for VR environments and digital avatars via modern neural rendering. Designed and co-developed components across the full vertical stack (artist tools, game engine integration, evaluation pipelines, core rendering systems), and mentored 10+ engineers and researchers in real-time rendering.
- Shipped **Meta Horizon Hyperscape**, a VR app showcasing photorealistic environments via Gaussian splatting.
- Reduced ML training costs by several million dollars annually by optimizing CUDA kernels and training pipelines for large-scale 3D reconstruction of digital avatars.
- Scaled high-fidelity cloth simulation and rendering for digital humans, leading components on CPU (SIMD) and GPU (CUDA) as well as real-time neural ML deformers (Vulkan, PyTorch) running on VR devices.
- Built several cross-platform computer graphics libraries and tools used by dozens of internal teams (acceleration structures, Vulkan abstraction layer, GPU profiling, Gaussian splatting runtime).

Software Engineer

Zurich, Switzerland

DAEDALEAN AG

September 2020 - August 2021

- Developed reliable autonomous piloting systems for aircraft using vision-based methods.
- Built a cloud-based rendering and simulation service generating large-scale multi-view aerial synthetic data for multiple internal teams.
- Improved the detect-and-avoid system by scaling training infrastructure, implementing sophisticated data augmentation (TensorFlow), and safety-critical testing.

Research Assistant

Zurich, Switzerland

ETH ZÜRICH

May 2019 - August 2020

- Developed novel techniques for dimensionality reduction, topology, feature extraction and visualization of high-dimensional flows and dynamical systems, using nonlinear optimization, machine learning and variational calculus.

Rendering Engineer

Zurich, Switzerland

KAPANU AG

June 2017 - April 2019

- Shipped **IvoSmile**, a mobile AR try-on tool for dentists that aids in communicating dental prosthetic treatments.
- Developed an efficient mobile-first software architecture (iOS, Android), a cross-platform real-time rendering system (OpenGL), optimized performance and built several geometry processing algorithms for dentures. Conducted R&D on rendering, machine learning and physical lab design for capturing dental materials.

Publications

SqueezeMe: Mobile-Ready Distillation of Gaussian Full-Body Avatars

SIGGRAPH 2025

FORREST IANDOLA, STANISLAV PIDHORSKYI, IGOR SANTESTEBAN, DIVAM GUPTA, ANUJ PAHUJA, **NEMANJA**

August 2025

BARTOLOVIC, FRANK YU, EMANUEL GARBIN, TOMAS SIMON, SHUNSUKE SAITO

- Developed a method to animate and render multiple digital human avatars in real-time on commodity VR hardware.

Hardware-Rasterized Ray-Based Gaussian Splatting

(**Highlight**) CVPR 2025

SAMUEL ROTA BULÒ, **NEMANJA BARTOLOVIC**, LORENZO PORZI, PETER KONTSCIEDER

June 2025

- Co-developed a novel rasterization-based Gaussian splatting algorithm that minimizes projection errors in high field-of-view settings like VR rendering.

Phase Space Projection of Dynamical Systems

(**Honorable Mention**) EuroVis 2020

NEMANJA BARTOLOVIC, MARKUS GROSS, TOBIAS GÜNTHER

May 2020

- Developed a flow and curvature-preserving dimensionality reduction technique for trajectories in high-dimensional dynamical system phase spaces.

PhysicsForests: Real-Time Fluid Simulation Using Machine Learning

SIGGRAPH 2017 Real-Time Live!

LUBOR LADICKY, SOHYEON JEONG, **NEMANJA BARTOLOVIC**, MARC POLLEFEYS, MARKUS GROSS

August 2017

- Implemented the rendering techniques (developed during my Master's thesis) used to visualize the fluid simulations, and contributed improvements to the core simulation framework. [Demo](#)

Talks

Gaussian Splatting: Ready for Standardization?

SIGGRAPH 2025

August 2025

- Described the state of Gaussian splatting at Meta and outlined practical forward-looking file format considerations.

Education

Master in Computer Science

Zurich, Switzerland

ETH ZÜRICH

September 2015 - April 2017

- Visual computing/computer graphics track. [Thesis](#) on real-time fluid rendering and real-time neural network style transfer.

Bachelor in Computer Science

Nis, Serbia

FACULTY OF ELECTRONIC ENGINEERING, UNIVERSITY OF NIS

October 2011 - July 2015

- Software engineering and informatics; Bachelor's thesis in computer graphics.

Personal Projects

Crisp

Personal Interest

RENDERING PROJECT

January 2016 - Present

- A personal renderer for experimenting with new ideas and algorithms in the domain of real-time rendering (rasterization and ray tracing) with a Vulkan backend. Built and maintained in my free time. [Project Page](#)

Technical Skills

Languages & APIs C++, Vulkan, OpenGL, Python, C, MATLAB, CUDA, SIMD

Tools VS Code, Visual Studio, RenderDoc, Nsight, Git, Mercurial, CMake, Bazel, Buck

General Software architecture, modern C++, optimization

Visual Computing Real-time rendering, ray tracing, geometry processing, neural rendering, physically-based simulation

Patents

2023	Method for Aligning a 3D Dentition Model to a Face Image (US 11,544,861)	Ivoclar Vivadent AG
2022	Computer-Implemented Method for Modifying a Digital 3D Dentition Model (US 11,344,392)	Ivoclar Vivadent AG
2022	Dental Model Attributes Transfer (US 11,244,519)	Ivoclar Vivadent AG
2022	Rendering a Dental Model in an Image (US 11,213,374)	Ivoclar Vivadent AG
2021	Rendering of Dental Models (US 11,083,552)	Ivoclar Vivadent AG
2021	Dental Design Transfer (US 10,980,621)	Ivoclar Vivadent AG